

# Quick Check

Lesson 5: Who Profits from Plant Wastes? Community Uses, Alkaloid Risks and the Final Explanation

*7 multiple-choice questions • choose A, B, C or D*

Sub-Strand 2.3: Excretion in Plants | Lesson 5

**Tea leaves from Kericho contain caffeine and miraa (khat) contains cathinone. Both are alkaloids. Which statement about alkaloids is correct?**

- A** Some alkaloids are useful in small controlled doses but harmful when abused
- B** All alkaloids are harmful and should never be used
- C** Alkaloids are harmless and cannot cause addiction
- D** Only alkaloids from tobacco and miraa can harm people

**A farmer cuts the stem of a Euphorbia hedge and thick milky latex oozes out and later hardens. Which excretory process released the latex?**

**A** Organ fall

**B** Exudation

**C** Transpiration

**D** Guttation

**In Nakuru, tanners soak animal hides in a solution made from the bark of Acacia or wattle trees. Which plant excretory product in the bark makes this possible, and what is it used for?**

- A** Pyrethrum, used to make insecticide
- B** Quinine, used to treat malaria
- C** Tannins, used for tanning hides into leather
- D** Caffeine, used to flavour drinks

**Early in the morning, droplets of liquid water are seen along the edges of arrowroot (taro) leaves. Which statement correctly contrasts guttation with transpiration?**

- A** Guttation loses water vapour through stomata, while transpiration loses liquid droplets through hydathodes
- B** Guttation and transpiration are the same process and both happen only at night
- C** Guttation removes sticky resin from cut stems, while transpiration removes fallen leaves
- D** Guttation loses liquid water droplets through hydathodes at leaf edges, while transpiration loses water vapour mainly through stomata

**A learner asks, 'Why does a plant not need a kidney?' Which answer is the best scientific reasoning?**

- A** Plants produce very toxic wastes that must be kept inside the cells at all times
- B** Plants absorb all their wastes from the soil and release them through the roots
- C** Plants have low metabolic rates and less toxic wastes, recycle CO<sub>2</sub> and O<sub>2</sub> between photosynthesis and respiration, and store other wastes in dead tissues
- D** Plants do not produce any wastes

A group interviewed 12 households in their community. 7 of the 12 households said they use neem leaves to treat malaria or skin problems. What percentage of the households use neem?

A 7%

B 70%

C 41.7%

D 58.3%

**In Lesson 1, a learner drew a model showing that plants have no wastes. After Lessons 1-5, which is the correct revised idea for the final model?**

- A** Plants have no wastes, so nothing needs to be removed
- B** Plants produce wastes, but these have no use to people
- C** Plants produce wastes, but all of them are removed through kidneys in the roots
- D** Plants produce wastes, but they remove or store them in different ways, such as exudation, guttation, deposition and organ fall

# All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).